

AYMANE AIT DADS

AI Developer Intern - Generative AI & RAG | LLM Fine-Tuning · Knowledge Base Integration · Business Process Automation

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT - directly demonstrating the LLM engineering depth required for enterprise-class Generative AI systems. Reduced manual network-analysis time by 60% at Orange Maroc by automating feature engineering pipelines across 100K+ records, connecting model outputs to actionable business recommendations. Brings hands-on experience in RAG architecture, LLM orchestration, agentic pipelines, prompt engineering, and knowledge base integration - the exact stack needed to build proposal-automation and internal-knowledge retrieval products at CGI. Aims to deploy production-grade Generative AI services that connect LLMs with organizational knowledge bases, accelerating enterprise business process automation.

CORE COMPETENCIES

Generative AI | RAG Architecture | LLM Orchestration | Business Process Automation | Knowledge Base Integration | Prompt Engineering | Agentic Pipelines | Enterprise AI Systems

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Automated feature engineering and model evaluation pipelines on **100K+ fiber-optic network records**, reducing manual analysis time by ~60% and delivering outputs directly to the network optimization team.
- Built a **clustering pipeline** mapping customer performance profiles to 5 commercial service tiers using Random Forest and K-Means, translating model outputs into actionable maintenance recommendations for stakeholders.
- Developed and presented stakeholder-facing reports translating complex model results into **business process recommendations**, bridging data science outputs and operational decision-making.
- Engineered end-to-end data pipeline ingesting upload speed, latency, and jitter signals to classify network quality across **thousands of nodes** using Python, Scikit-learn, Pandas, and Power BI.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (Jun 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in an ongoing competition by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation), building a diagnostic evaluation pipeline that revealed reasoning over-expansion - not incorrect reasoning - as the primary failure mode, and constructed custom cipher parsers and 8-bit operation solvers to produce verified chain-of-thought training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection - NTIRE

EURECOM ImSecu Course + NTIRE 2026 @ CVPR CodaBench · 2025 · Ranked 1st in Class

2026 @ CVPR

- Ranked **1st on the EURECOM private class leaderboard** (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head and dual learning-rate strategy (2e-6 backbone, 5e-4 head) trained on 250K images across 25+ generator types; also submitted independently to the NTIRE 2026 @ CVPR CodaBench international benchmark.
- Discovered that **1-epoch training (0.793 AUC) outperformed 4-epoch (0.767 AUC)** for out-of-distribution generalization, and built a 3-model ensemble with 10-view TTA (flips, crops, blur, resizes) weighted by validation AUC.

PyTorch · OpenCLIP · ViT · FP16 · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Web Applications, Image Security, Statistics

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Generative AI: LoRA / QLoRA | SFTTrainer | vLLM | RAG | LangChain | Agentic Pipelines | Prompt Engineering

ML / DL & NLP: PyTorch | Hugging Face Transformers | Scikit-learn | OpenCLIP | TRL | PEFT | chain-of-thought

Data Engineering & MLOps: Python | Pandas | SQL | Docker | REST APIs | Supabase | Power BI | Git